

# SpaceX Falcon 9 Landing Prediction

Applied Data Science Capstone | Final Report

End-to-end workflow: data collection -> wrangling -> EDA -> geospatial analytics -> dashboard -> classification models

90

cleaned launch records

80

feature columns for ML

83.3%

best test accuracy

GitHub repository: <https://github.com/Nikita18365/Coursera-Applied-Data-Science-Capstone>

Nikita Garamov

# Executive Summary

## Business question

Can launch, payload, orbit, site and booster attributes help predict whether Falcon 9 first stage will land successfully?

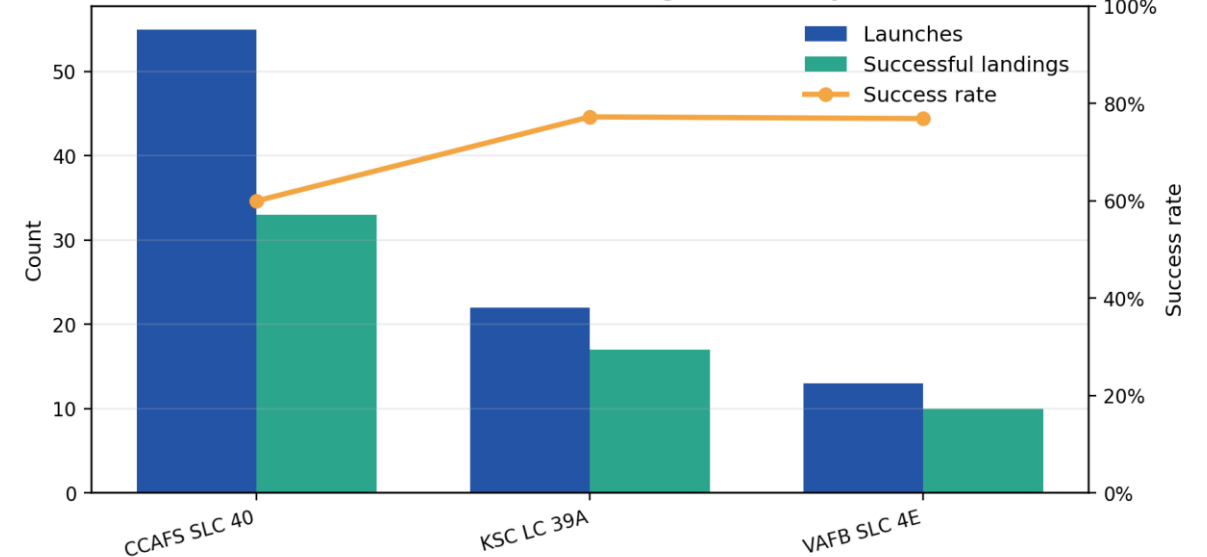
## Core finding

Success improved strongly over time: yearly landing success moved from 0% in early years to above 80% in 2017, 2019 and 2020.

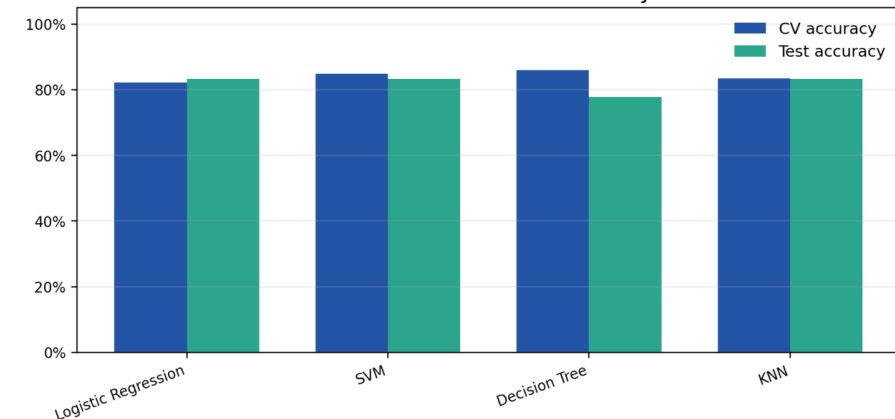
## Predictive result

Logistic Regression, SVM and KNN reached 83.3% accuracy on the held-out test set; SVM was selected by combined CV/test performance.

Launch volume and landing success by site



Classification model accuracy



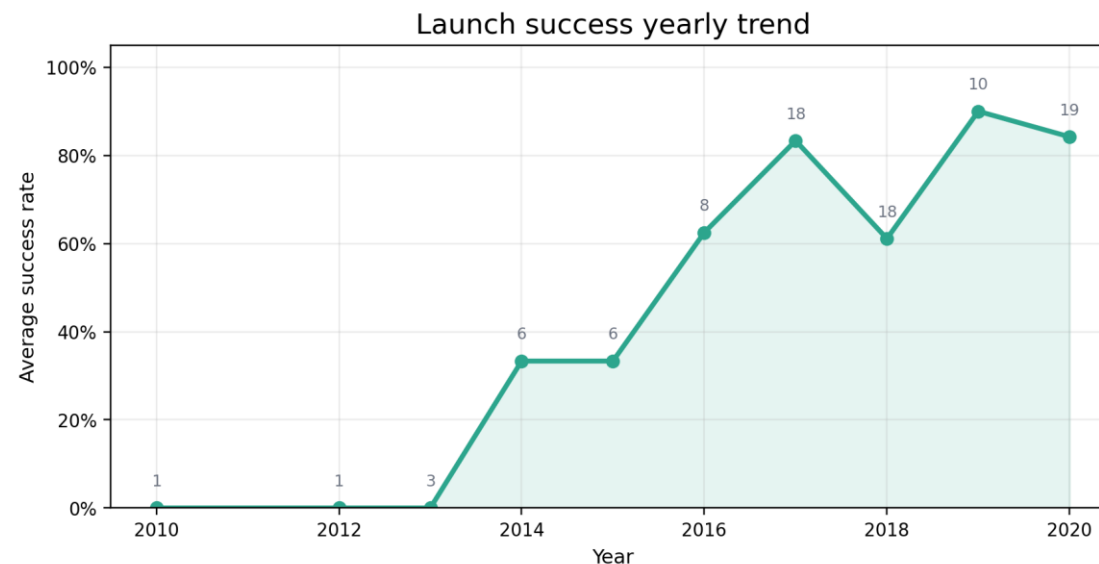
# Introduction: from raw launch data to a landing predictor

Reusable first-stage landing is a critical operational milestone for Falcon 9 launches.

The project treats landing outcome as a supervised classification problem.

The workflow integrates structured API data, web-scraped launch tables, SQL analysis, geospatial exploration, interactive dashboarding and machine learning.

The final output is a reproducible data science pipeline and an interpretable set of launch-success drivers.



*Landing outcomes improved materially after 2015, motivating predictive analysis.*

**Class**

target variable

**1**

success class

**0**

failure class

# Repository and deliverables

GitHub URL

<https://github.com/Nikita18365/Coursera-Applied-Data-Science-Capstone>

| Folder                        | Deliverable                                          |
|-------------------------------|------------------------------------------------------|
| 1. Data collection            | SpaceX API notebook, web-scraping notebook, raw CSVs |
| 2. Data wrangling             | Cleaned dataset with landing outcome label           |
| 3. EDA with SQL               | SQLite database and SQL notebook                     |
| 4. EDA with Pandas/Matplotlib | Visualization notebook and feature CSV               |
| 5. Folium analytics           | Interactive map notebook                             |
| 6. Plotly Dash                | Dashboard application and findings                   |
| 7. Predictive analysis        | Classification notebook                              |

## Submission artifact

The completed slide report is exported to PDF as required by the course rubric.

## Review focus

The deck is more technical than an executive presentation: it includes methodology, code-driven outputs, SQL results, maps, dashboard results and model evaluation.

## Innovation layer

Beyond the template, the report adds an end-to-end pipeline view, model comparison, operational interpretation and suggested next-step features.

# Methodology overview



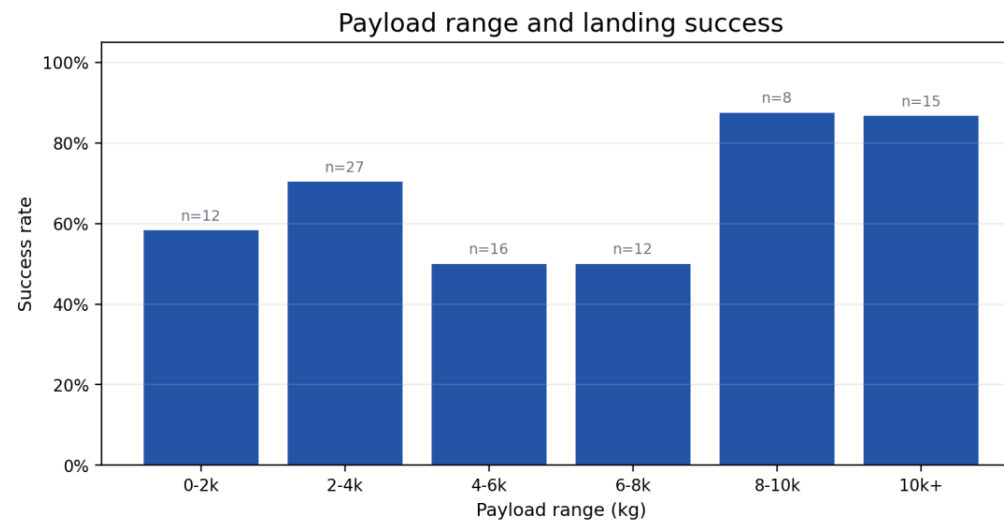
Collected launch data using SpaceX endpoints and Wikipedia launch tables.

Created a binary landing-success target and selected predictive features.

Explored relationships across site, orbit, payload, booster configuration and year.

Used interactive tools to inspect geography and dashboard-level patterns.

Compared tuned classifiers using cross-validation and a held-out test split.



*Payload bins offer a simple operational view of the landing-success distribution.*

# Data collection methodology: SpaceX REST API



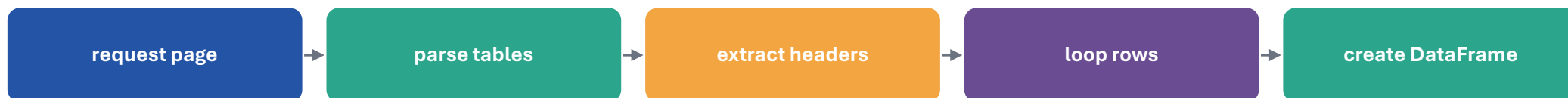
Requested historical SpaceX launch records and enriched them with launchpad, payload, rocket and core metadata.  
Converted nested JSON response fields into tabular columns.  
Output dataset\_part\_1.csv contained 90 records and 17 columns before target engineering.  
Key fields: FlightNumber, Date, PayloadMass, Orbit, LaunchSite, Outcome, Booster/Block, reused hardware indicators, coordinates.

| Metric               | Value                            |
|----------------------|----------------------------------|
| API rows             | 90                               |
| Initial columns      | 17                               |
| Coordinates captured | Longitude, Latitude              |
| Main identifiers     | FlightNumber, Serial, LaunchSite |

## Repository evidence

Folder 1 includes the completed SpaceX API collection notebook and generated CSV.

# Data collection methodology: web scraping



Used BeautifulSoup to parse Falcon 9 launch tables from the web source used in the course lab.

Extracted flight number, launch site, payload, payload mass, orbit, customer, launch outcome, booster version and landing outcome.

The scraped table provided an external cross-check and richer descriptive launch history.

Output spacex\_web\_scraped.csv contained 121 rows and 11 columns.

| Field group  | Examples                               |
|--------------|----------------------------------------|
| Launch data  | Flight No., Date, Time, Launch site    |
| Mission data | Payload, payload mass, orbit, customer |
| Outcome data | Launch outcome, booster landing        |
| Booster data | Version Booster                        |

## Scraping insight

The scraped dataset shows landing outcome as text; this is useful for descriptive comparison but required cleaning before modeling.

# Data wrangling: target label and model-ready features

Checked missing values and data types; LandingPad had the highest missing percentage.

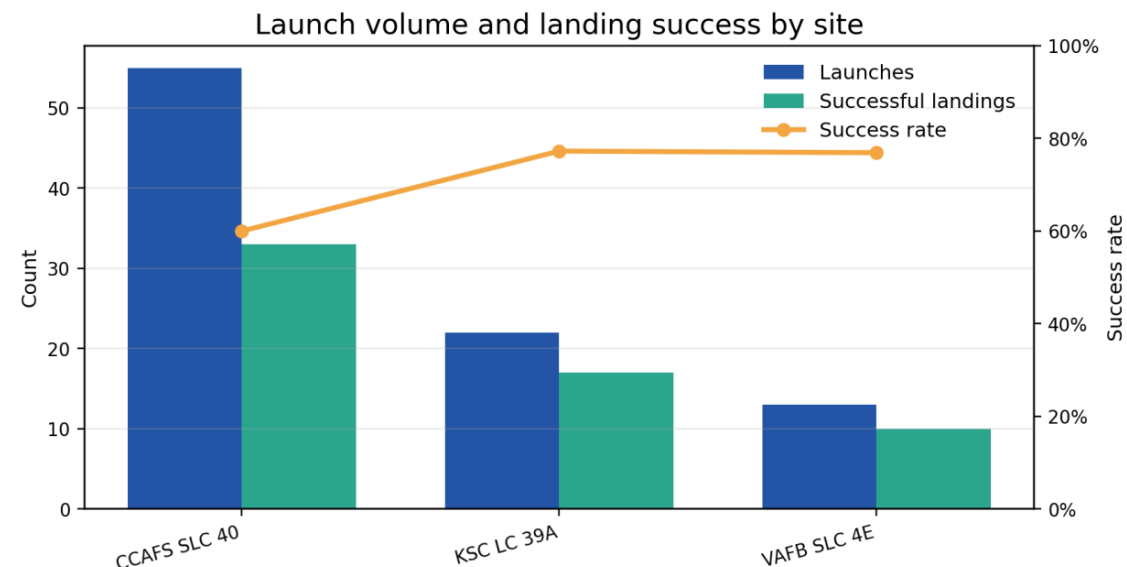
Imputed missing PayloadMass values using the mean value from available records.

Converted landing outcomes into a binary target: Class = 1 for successful landing, Class = 0 otherwise.

Created one-hot encoded features for orbit, launch site, landing pad and booster serial.

Final model input table contained 90 rows and 80 feature columns.

| Dataset            | Rows | Columns |
|--------------------|------|---------|
| dataset_part_1.csv | 90   | 17      |
| dataset_part_2.csv | 90   | 18      |
| dataset_part_3.csv | 90   | 80      |



## Target distribution

The cleaned dataset contains 60 successful and 30 failed first-stage landing outcomes.

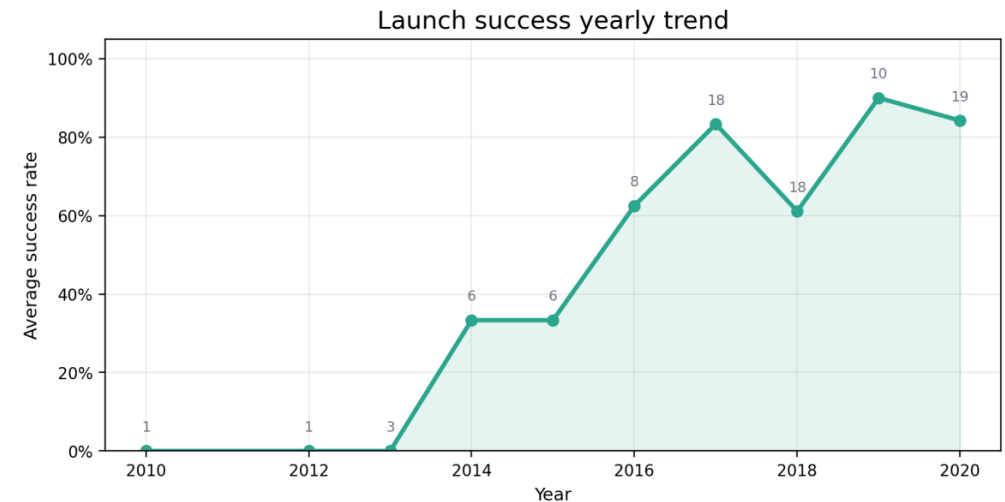
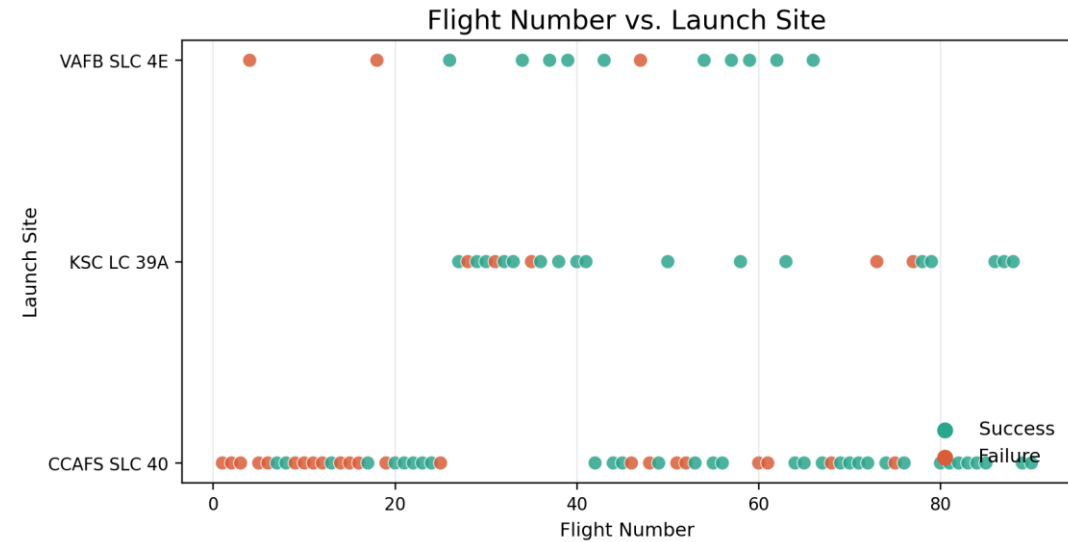
# EDA methodology: Pandas and Matplotlib

Scatter plots were used to inspect individual launch records across flight number, payload, site and orbit.

Bar charts summarized aggregated success rates by orbit and payload range.

Line charts showed temporal movement in average yearly success rate.

The EDA goal was to identify visual patterns before building classification models.



## Why these charts?

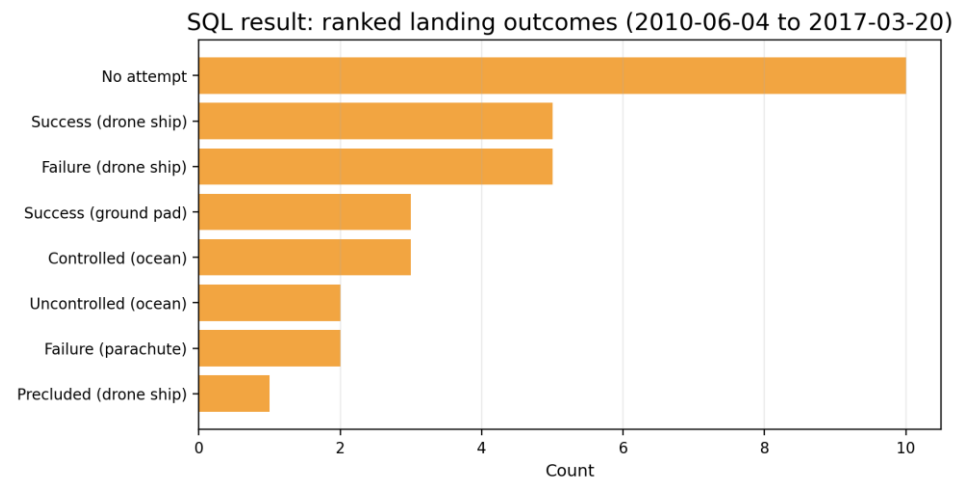
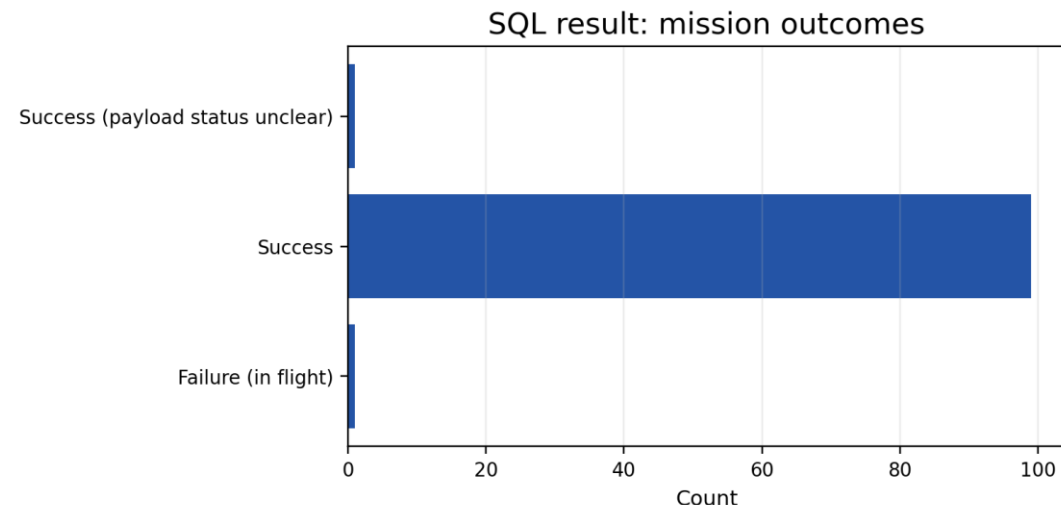
The plots connect business drivers such as launch site, payload mass, orbit type and learning-over-time to the final landing outcome.

# EDA methodology: SQL queries

Loaded launch data into SQLite as SPACEXTABLE.  
Used SELECT DISTINCT, WHERE, LIKE, SUM, AVG, MIN, GROUP BY and ORDER BY to answer structured questions.  
SQL was used for reproducible summaries: site names, CCA records, NASA payload, booster payload statistics, landing dates and outcome rankings.  
SQL results complement visual EDA by making aggregate answers explicit.

## SQL table

SPACEXTABLE: Date, Time, Booster\_Version, Launch\_Site, Payload, PAYLOAD\_MASS\_KG\_, Orbit, Customer, Mission\_Outcome, Landing\_Outcome.



# Interactive visual analytics methodology

Folium map objects: Circle, Marker, MarkerCluster, DivIcon distance labels and PolyLine distance connectors.

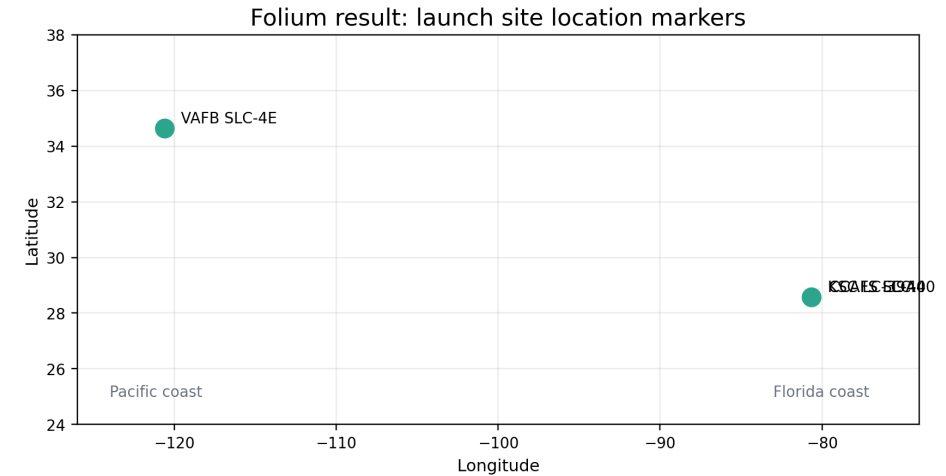
Dash input components: launch-site dropdown and payload range slider.

Dash output graphs: pie chart for success/failure composition and scatter plot for payload vs. landing outcome.

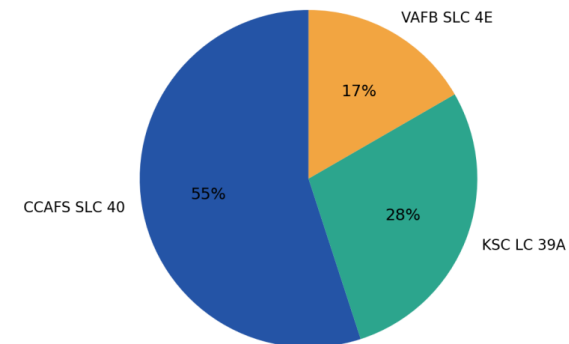
Interactivity supports questions that are hard to answer from static tables alone.

## User-controlled analysis

The dashboard lets a reviewer move from all sites to one launch site, then inspect selected payload ranges interactively.



Dash result: successful launches by site



# Predictive analysis methodology



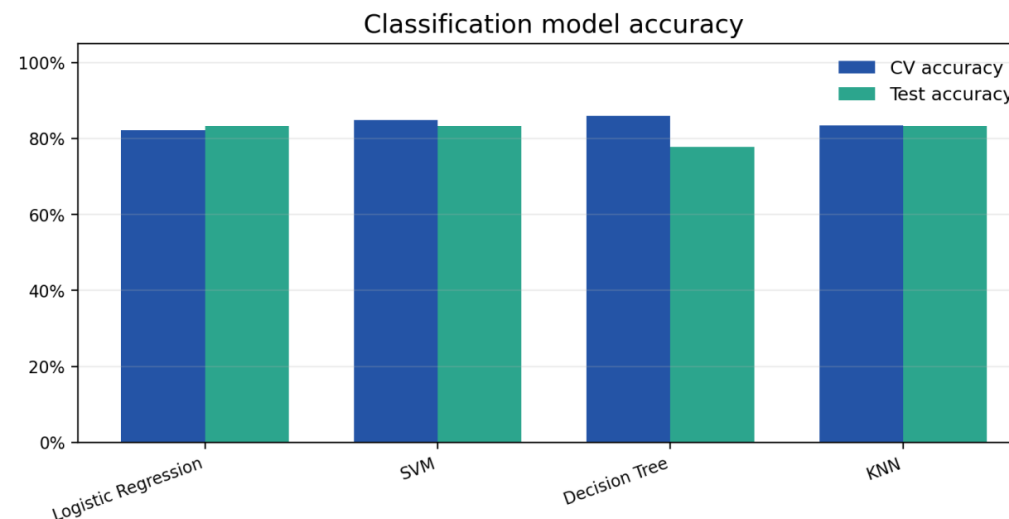
Target: first-stage landing outcome, Class = 1 for success and Class = 0 for failure.

The feature matrix was standardized before fitting models.

Train/test split: 80% training and 20% testing with random\_state = 2.

Classifiers tuned: Logistic Regression, SVM, Decision Tree and KNN.

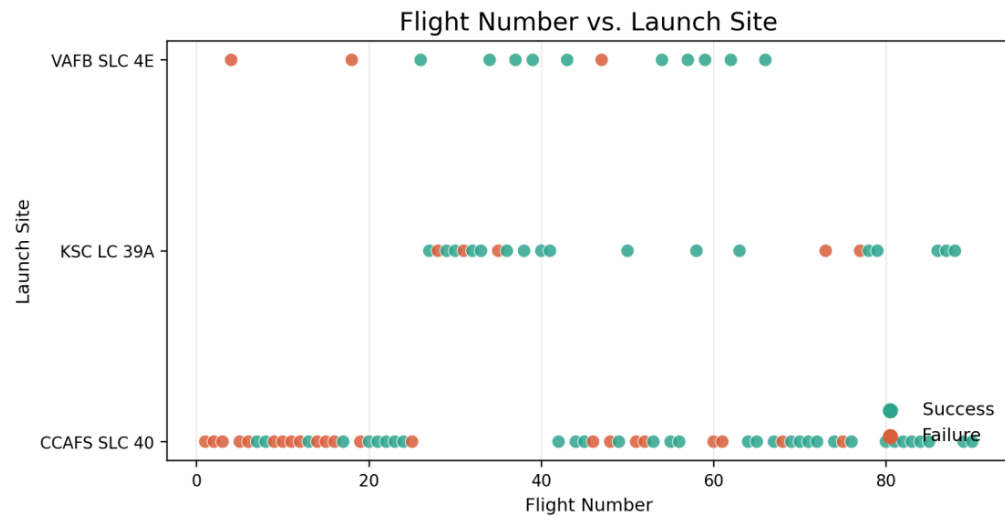
Model selection considered cross-validation performance and test-set accuracy.



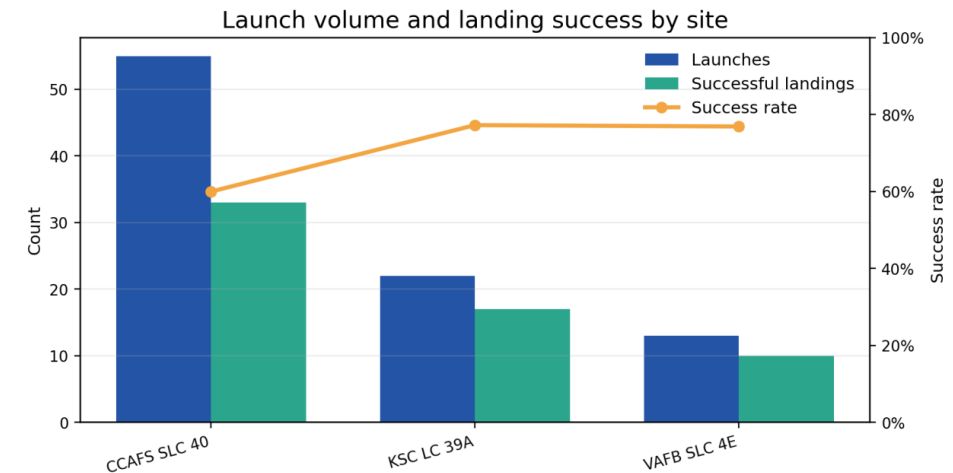
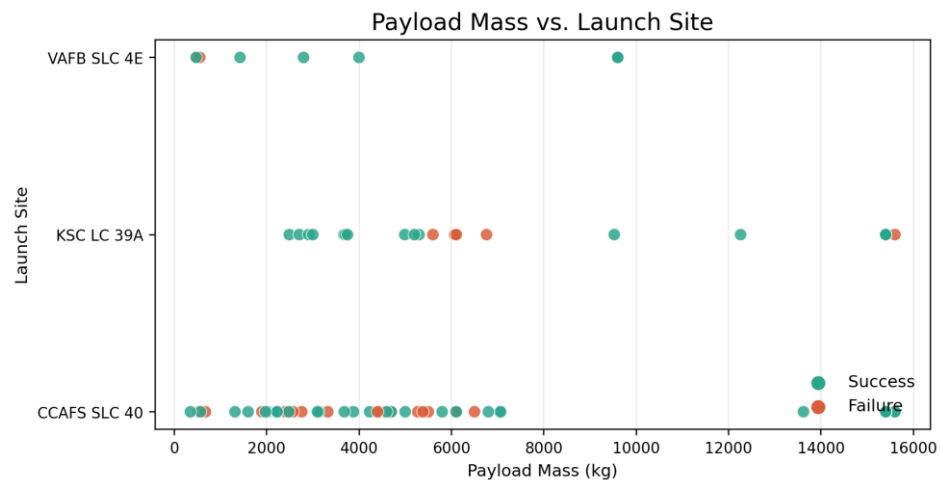
## Evaluation metric

Accuracy was used to compare models; the confusion matrix was used to inspect error type.

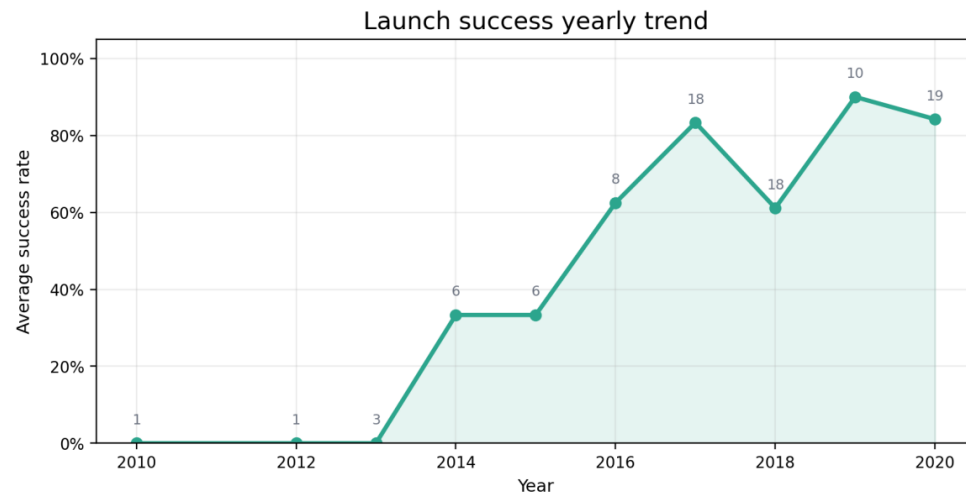
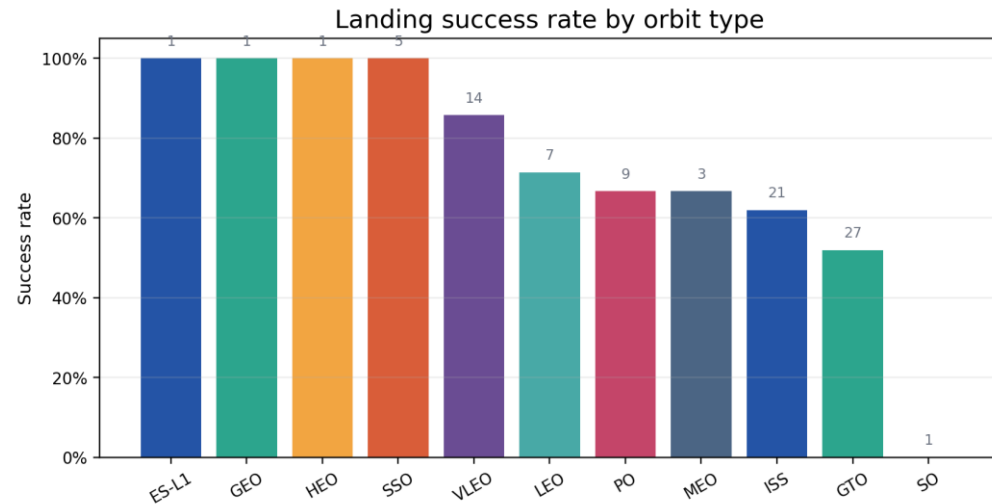
# EDA results: launch site and payload patterns



CCAFS SLC 40 has the largest launch volume and the largest absolute count of successful landings in the 90-row dataset. KSC LC 39A and VAFB SLC 4E have higher success rates despite fewer launches. Payload mass alone does not separate success and failure completely; site and time effects still matter.



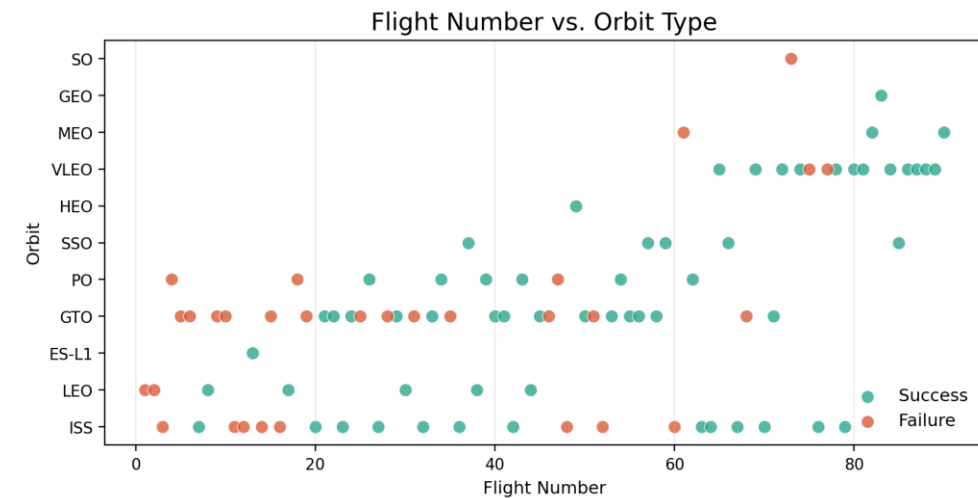
# EDA results: orbit type and yearly learning curve



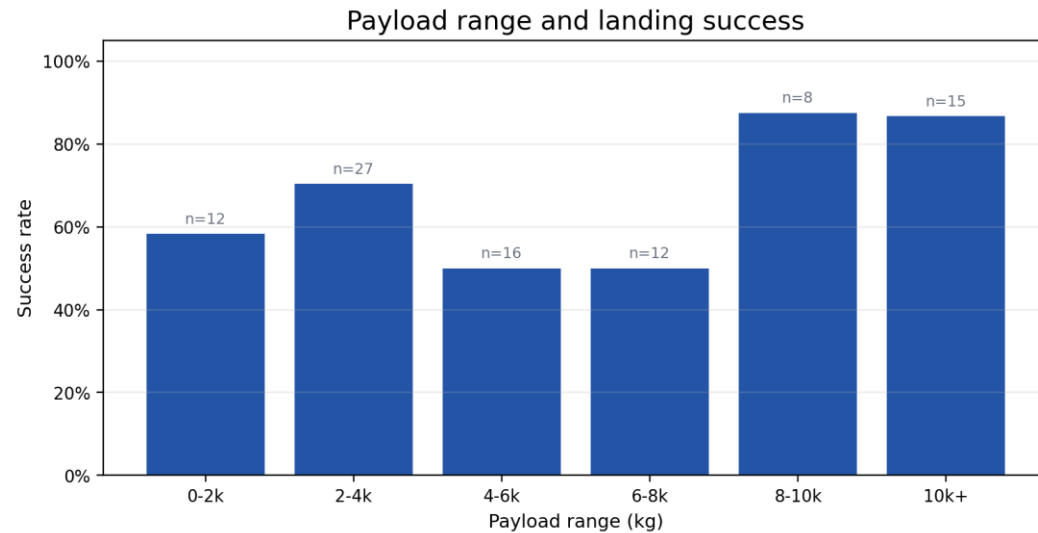
High success rates appear in SSO, VLEO and several small-sample orbit categories.

GTO and ISS contain more records and show mixed outcomes, so they are more informative for modeling.

The time trend shows a strong improvement after the early launch period.



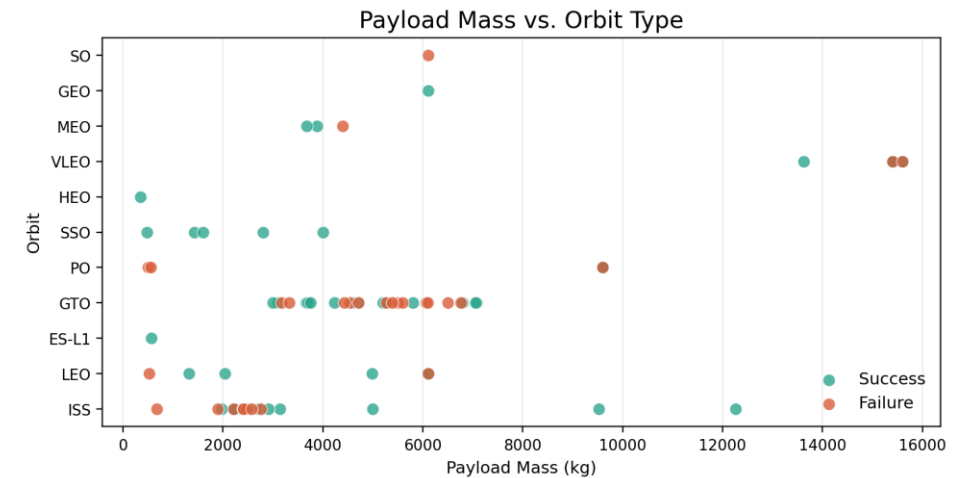
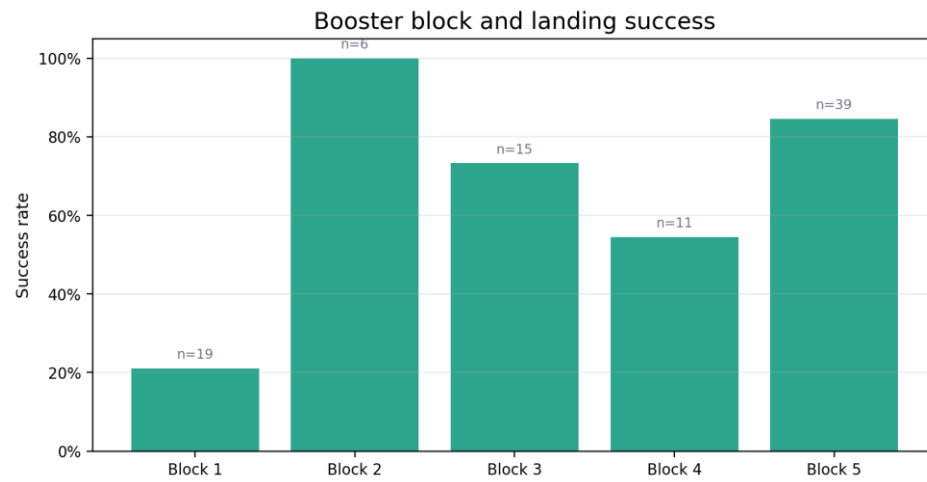
# EDA results: payload range and booster generation



Payload ranges above 8,000 kg show high success rates in the cleaned dataset, but sample sizes vary.

Earlier booster blocks have lower success in the course dataset; later blocks show higher landing reliability.

This supports using hardware-generation variables as predictive features.



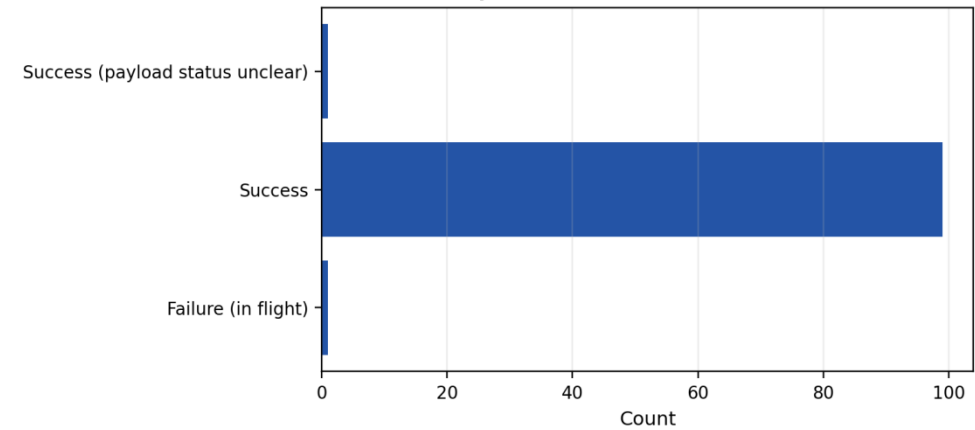
# SQL results: launch sites, NASA payload and mission outcomes

| Query                               | Result                                             |
|-------------------------------------|----------------------------------------------------|
| Distinct launch sites               | CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40 |
| Total payload by NASA customers     | 107,010 kg                                         |
| Average payload for F9 v1.1         | 2,534.67 kg                                        |
| First successful ground-pad landing | 2015-12-22                                         |

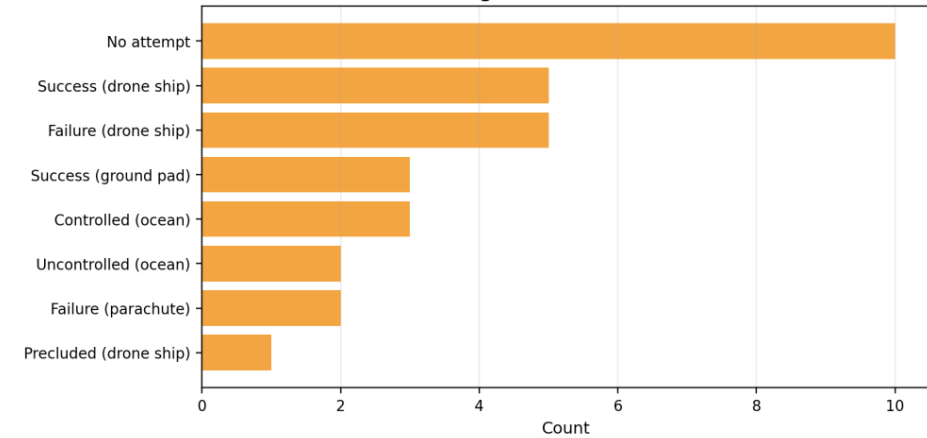
## Interpretation

SQL confirms that most mission outcomes were successful, while landing outcomes were more varied during the early learning period.

SQL result: mission outcomes



SQL result: ranked landing outcomes (2010-06-04 to 2017-03-20)



# SQL results: booster payload and selected historical queries

| SQL question                                               | Answer                                |
|------------------------------------------------------------|---------------------------------------|
| Successful drone-ship landings with 4,000-6,000 kg payload | F9 FT B1022, B1026, B1021.2, B1031.2  |
| Boosters carrying maximum payload                          | Multiple F9 B5 boosters at 15,600 kg  |
| Failed drone-ship landings in 2015                         | 2015-01-10 B1012 and 2015-04-14 B1015 |
| Top ranked landing outcome by count before 2017-03-20      | No attempt: 10 records                |

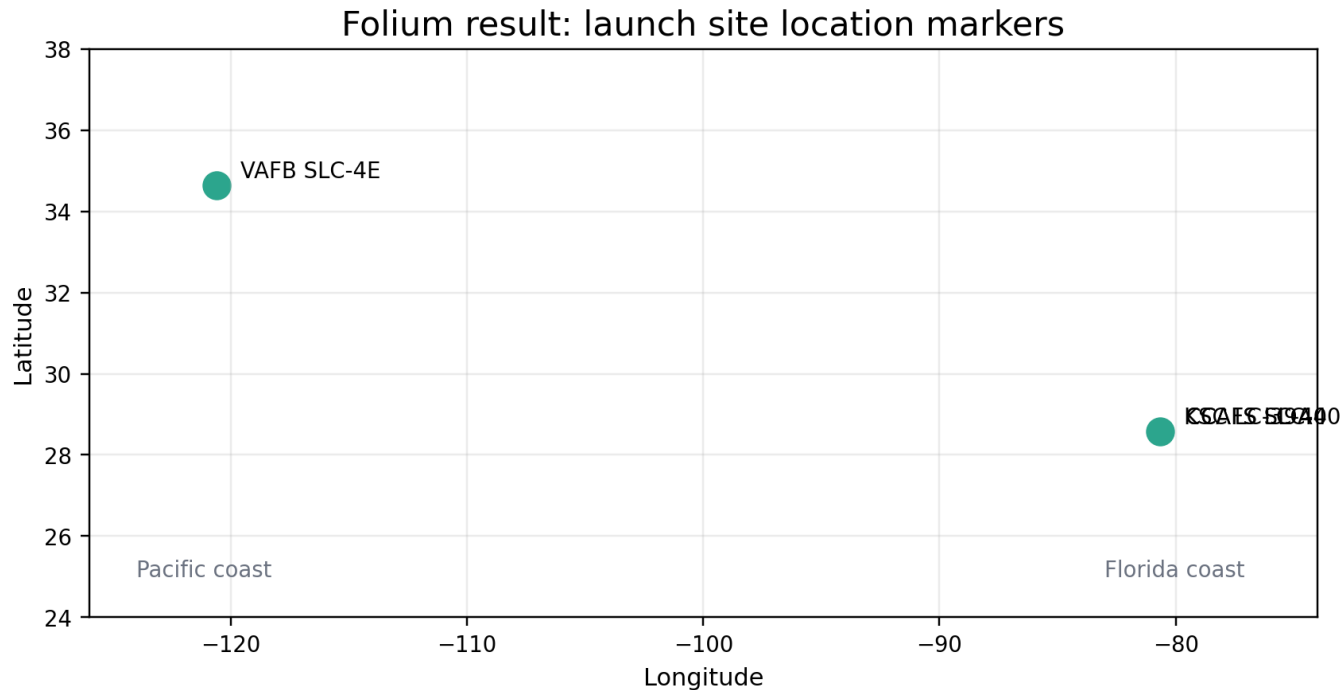
## Why this matters

The SQL tasks demonstrate reproducible filtering across date, payload, launch site, booster version and landing-outcome dimensions.

## Operational lens

Payload extremes are mainly associated with F9 B5 entries, consistent with later-generation booster capability.

# Folium results: launch site location markers



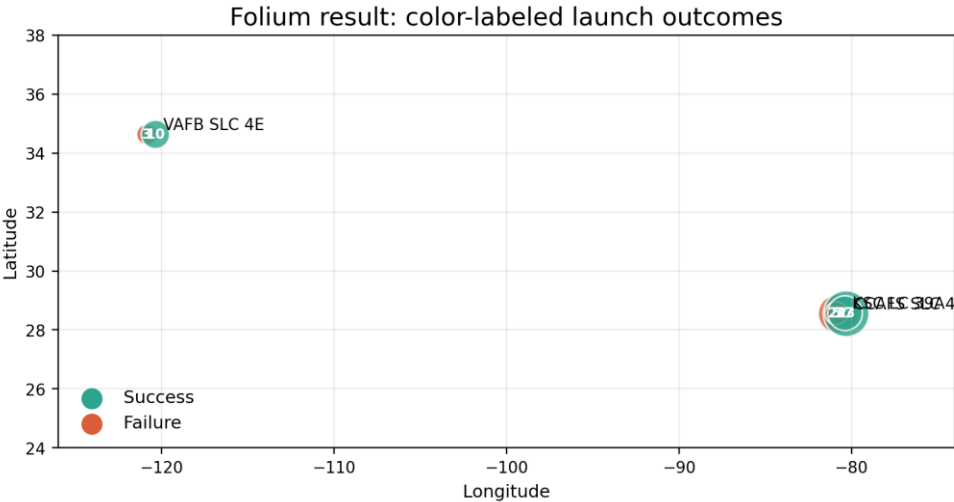
All launch sites are north of the Equator; they are not equatorial locations.

Each site is located very close to a coast, which supports safer launch trajectories over water. Florida sites cluster together; VAFB SLC-4E represents the west-coast launch geography.

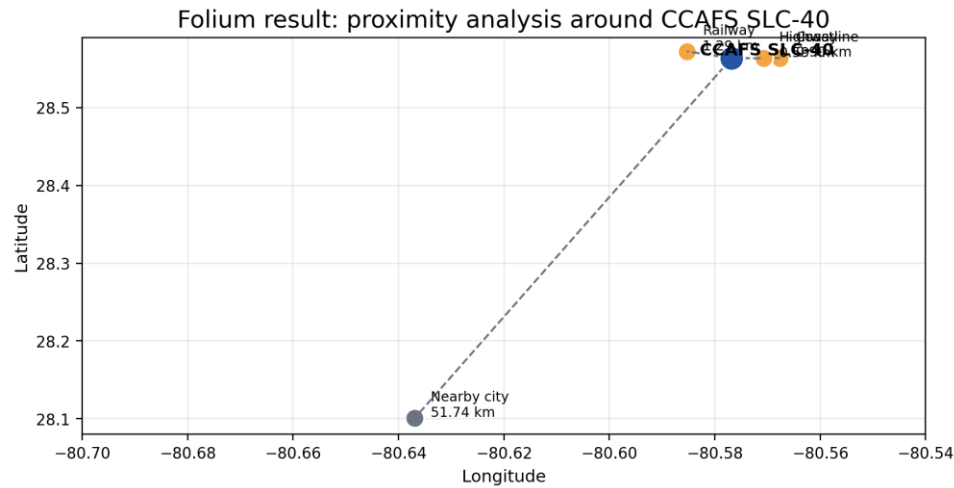
## Folium objects used

Circle and Marker objects were added for launch-site locations on an interactive map.

# Folium results: outcomes and launch-site proximities



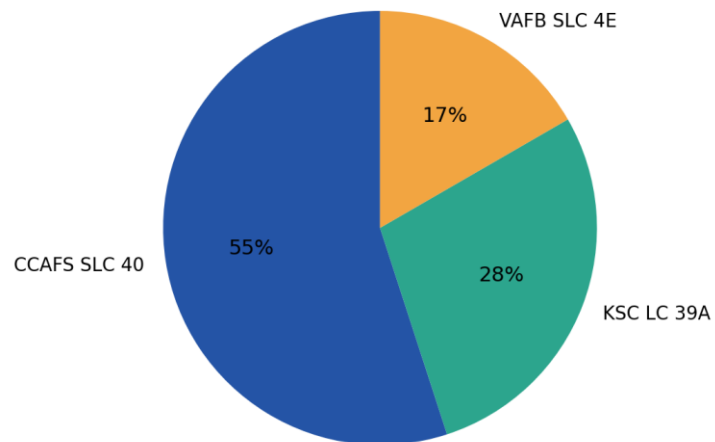
MarkerCluster groups multiple launch records at each location and makes repeated launch outcomes interpretable. For CCAFS SLC-40, the selected coastline, highway and railway points are close, while the nearby city point is much farther away. This pattern is operationally logical: launch sites require transport access but should reduce risk to urban populations.



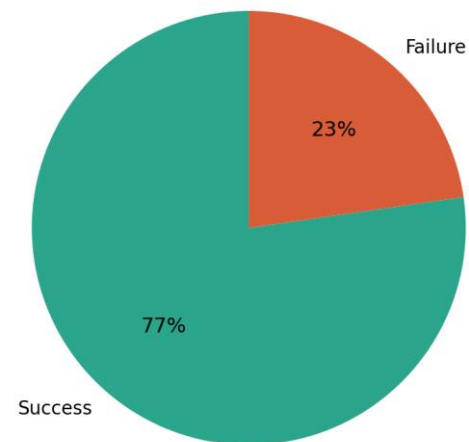
| Proximity   | Distance |
|-------------|----------|
| Coastline   | 0.90 km  |
| Highway     | 0.59 km  |
| Railway     | 1.29 km  |
| Nearby city | 51.74 km |

# Plotly Dash results: launch-site dropdown and success pie charts

Dash result: successful launches by site

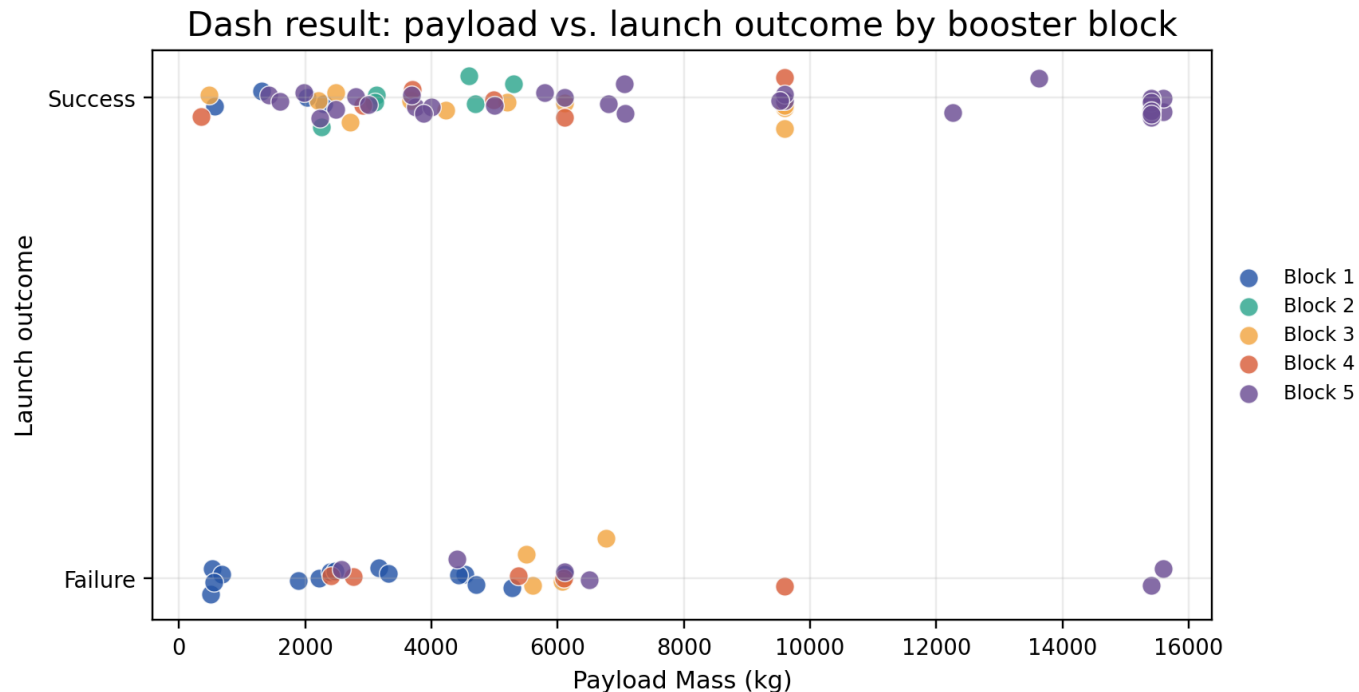


Dash result: KSC LC-39A success vs. failure



The dropdown toggles between all sites and one selected launch site.  
The all-sites pie uses launch site as category and success count as value.  
For a selected site, the pie switches to success vs. failure composition.  
This lets a reviewer quickly compare volume and reliability.

# Plotly Dash results: payload slider and scatter chart

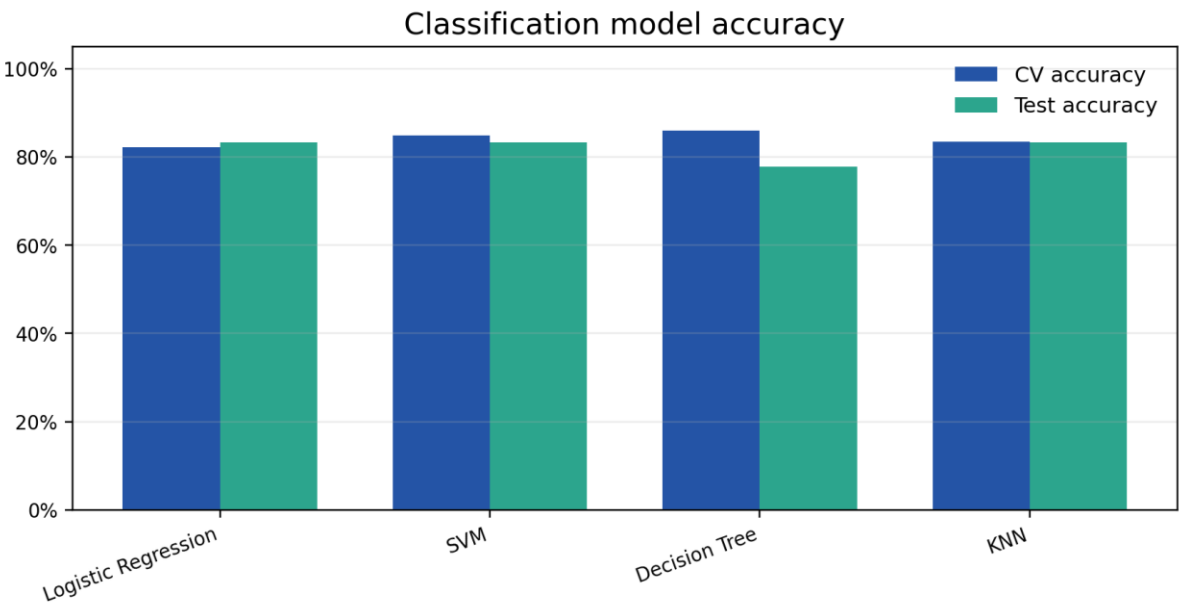


The range slider filters records by payload mass. The scatter plot compares payload mass against binary landing outcome. Color encoding shows booster block/category, making hardware-generation patterns visible. Payload is informative but not sufficient alone; booster and site context remain important.

## Dashboard insight

Later booster generations show stronger outcome stability in the course dataset.

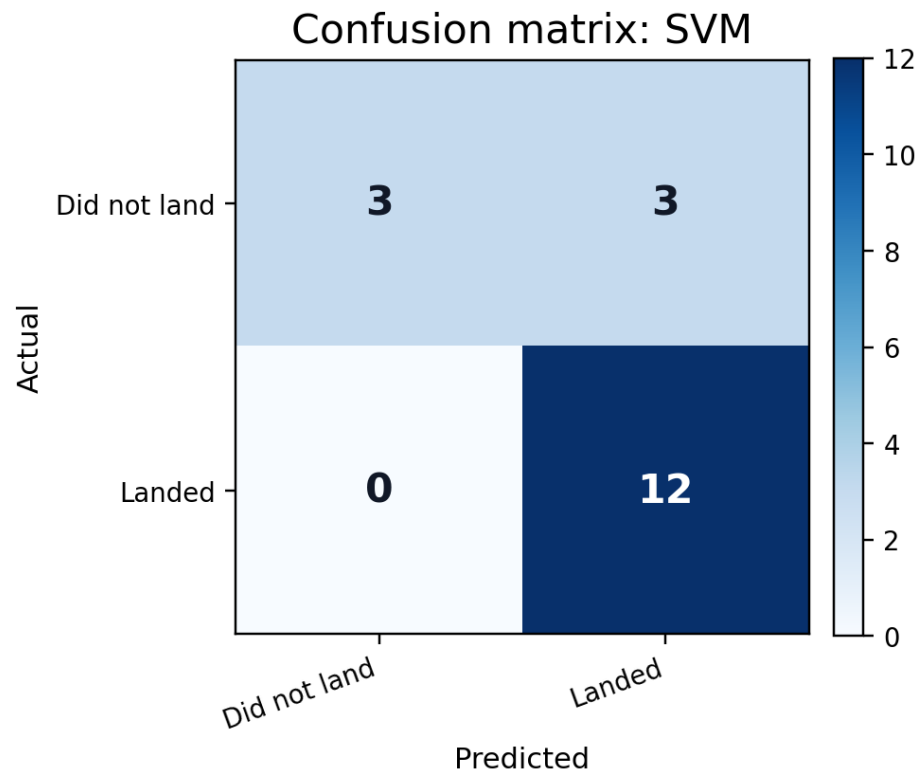
# Predictive analysis results: classification accuracy



| Model               | CV accuracy | Test accuracy |
|---------------------|-------------|---------------|
| Logistic Regression | 82.1%       | 83.3%         |
| SVM                 | 84.8%       | 83.3%         |
| Decision Tree       | 85.9%       | 77.8%         |
| KNN                 | 83.4%       | 83.3%         |

Three models tie at 83.3% test accuracy, but SVM has stronger cross-validation accuracy than Logistic Regression and KNN. Decision Tree achieves the highest CV score but lower test accuracy, suggesting less stable generalization on the small test set. The selected best method is SVM using the combined CV/test interpretation.

# Predictive analysis results: confusion matrix



## Best selected model

SVM reached 83.3% test accuracy and 84.8% cross-validation accuracy.

## Error profile

The selected model correctly predicted all 12 successful landings in the test split and missed 3 of 6 failed outcomes.

## Interpretation

The model is stronger at recognizing successful landings than failures, which is expected with a success-heavy target distribution.

# Conclusions

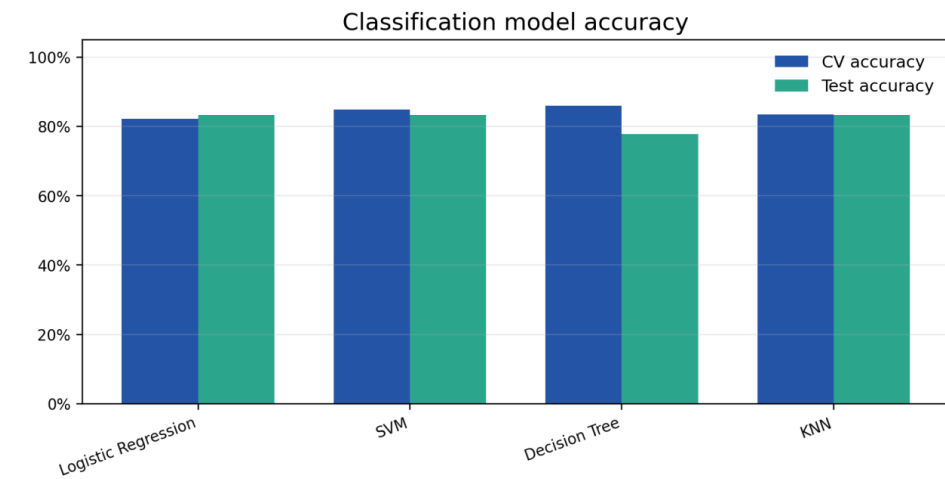
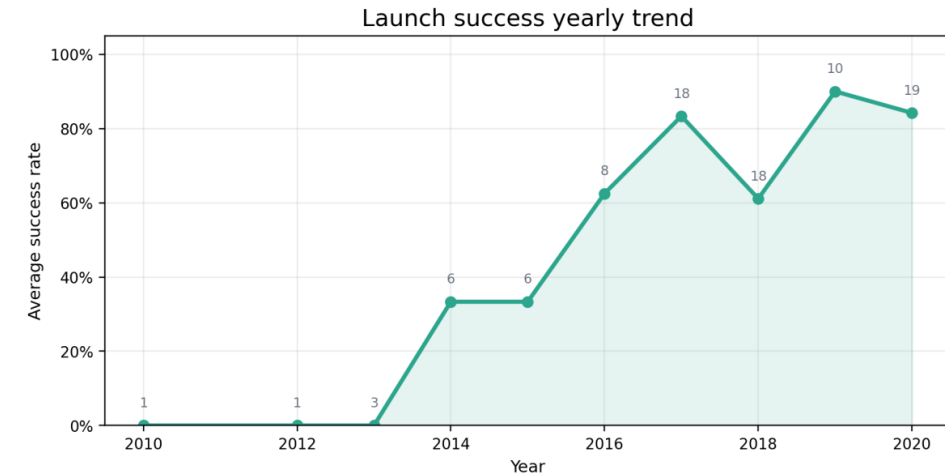
Launch outcomes improved sharply over time, reflecting operational learning and booster evolution.

Launch site, orbit type, booster generation and payload range all provide useful context for interpreting landing success.

CCAFS SLC 40 has the largest volume in the cleaned dataset, while KSC LC 39A and VAFB SLC 4E show stronger success rates.

Interactive Folium and Dash views make patterns easier to explain than tables alone.

A tuned SVM model achieved 83.3% held-out accuracy and provides a reasonable baseline for predicting first-stage landing success.



# Innovative ideas and next steps

Add weather, wind shear, sea-state and launch-window features to connect operations to landing probability.

Use rolling time-based validation instead of random split to better simulate future launches.

Add model explainability with permutation importance or SHAP to identify the most influential features.

Deploy the Dash app as a lightweight decision-support tool for exploring hypothetical payload/site scenarios.

Extend the analysis to post-2020 launches and compare Falcon 9 with Falcon Heavy missions.



## Beyond the template

The final report connects every lab output into one story: data engineering, exploration, interaction and prediction.

## Strategic value

The same workflow can be adapted to other launch providers, vehicle families or mission assurance problems.

# Appendix: completed course artifacts

| Required component           | Repository location                                                      |
|------------------------------|--------------------------------------------------------------------------|
| Data collection API notebook | 1. Data collection / jupyter-labs-spacex-data-collection-api.ipynb       |
| Web scraping notebook        | 1. Data collection / jupyter-labs-webscraping.ipynb                      |
| Data wrangling notebook      | 2. Data wrangling / labs-jupyter-spacex-Data wrangling.ipynb             |
| SQL EDA notebook             | 3. EDA with SQL / jupyter-labs-eda-sql-coursera_sqlite.ipynb             |
| EDA visualization notebook   | 4. EDA with Pandas and Matplotlib / edadataviz.ipynb                     |
| Folium notebook              | 5. Visual analytics with Folium / lab_jupyter_launch_site_location.ipynb |
| Dash app                     | 6. Plotly Dash / spacex-dash-app.py                                      |
| Predictive analysis notebook | 7. Predictive analysis / SpaceX_Machine Learning Prediction_Part_5.ipynb |

[GitHub: https://github.com/Nikita18365/Coursera-Applied-Data-Science-Capstone](https://github.com/Nikita18365/Coursera-Applied-Data-Science-Capstone)

# Appendix: key numeric outputs

| Site         | Launches | Success | Rate  |
|--------------|----------|---------|-------|
| CCAFS SLC 40 | 55       | 33      | 60.0% |
| KSC LC 39A   | 22       | 17      | 77.3% |
| VAFB SLC 4E  | 13       | 10      | 76.9% |

| Orbit | Success rate | Launches |
|-------|--------------|----------|
| SSO   | 100.0%       | 5        |
| VLEO  | 85.7%        | 14       |
| LEO   | 71.4%        | 7        |
| PO    | 66.7%        | 9        |
| ISS   | 61.9%        | 21       |
| GTO   | 51.9%        | 27       |

| Payload range | Success rate | Launches |
|---------------|--------------|----------|
| 0-2k          | 58.3%        | 12       |
| 2-4k          | 70.4%        | 27       |
| 4-6k          | 50.0%        | 16       |
| 6-8k          | 50.0%        | 12       |
| 8-10k         | 87.5%        | 8        |
| 10k+          | 86.7%        | 15       |

| Model              | Best test accuracy |
|--------------------|--------------------|
| LogReg / SVM / KNN | 83.3%              |
| Decision Tree      | 77.8%              |
| Selected model     | SVM                |

Tables are calculated from the supplied course datasets in the repository.